

Problem 14.2.9

Integrate $f(z)$ counterclockwise around the unit circle. Indicate whether Cauchy's integral theorem applies. Show the details.

$$f(z) = \exp(-z^2)$$

Solution

Problem 14.2.11

Integrate $f(z)$ counterclockwise around the unit circle. Indicate whether Cauchy's integral theorem applies. Show the details.

$$f(z) = \frac{1}{2z - 1}$$

Solution

Problem 14.2.13

Integrate $f(z)$ counterclockwise around the unit circle. Indicate whether Cauchy's integral theorem applies. Show the details.

$$f(z) = \frac{1}{z^4 - 1.1}$$

Solution

Problem 14.2.18

Integrate $f(z)$ counterclockwise around the unit circle. Indicate whether Cauchy's integral theorem applies. Show the details.

$$f(z) = \frac{1}{4z - 3}$$

Solution

Problem 14.2.22

Evaluate the integral. Does Cauchy's theorem apply? Show details.

$$\oint_C \operatorname{Re} z \, dz$$

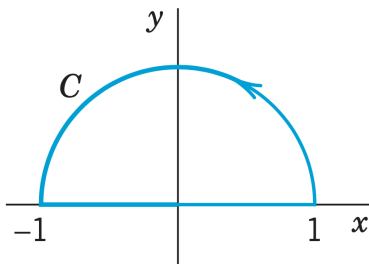


Figure 1: C

Solution

Problem 14.2.24

Evaluate the integral. Does Cauchy's theorem apply? Show details. (*Use partial fraction*)

$$\oint_C \operatorname{Re} z \, dz$$

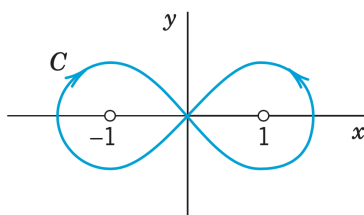


Figure 2: C

Solution

Problem 14.3.3

Integrate by Cauchy's formula counterclockwise around the circle.

$$\frac{z^2}{z^2 - 1}, \quad |z + i| = 1.4$$

Solution

Problem 14.3.11

Integrate counterclockwise.

$$\oint_C \frac{dz}{z^2 + 4}, \quad C : 4x^2 + (y - 2)^2 = 4$$

Solution**Problem 14.3.13**

Integrate counterclockwise.

$$\oint_C \frac{z+2}{z-2} dz, \quad C : |z-1| = 2$$

Solution**Problem 14.3.15**

Integrate counterclockwise. (compute it for two squares: one with vertices $\pm 2, \pm 2i$; the other one with vertices $\pm 4, \pm 4i$)

$$\oint_C \frac{\cosh(z^2 - \pi i)}{z - \pi i} dz, \quad C \text{ the boundary of the square with vertices } \pm 2, \pm 2, \pm 4i$$

Solution**Problem 14.4.1**

Integrate counterclockwise around the unit circle.

$$\oint_C \frac{\sin z}{z^4} dz$$

Solution**Problem 14.4.3**

Integrate counterclockwise around the unit circle.

$$\oint_C \frac{e^z}{z^n} dz, \quad n = 1, 2, \dots$$

Solution**Problem 14.4.6**

Integrate counterclockwise around the unit circle.

$$\oint_C \frac{dz}{(z-2i)^2(z-i/2)^2}$$

Solution**Problem 14.4.11**

Integrate. Show the details. *Hint.* Begin by sketching the contour. Why?

$$\oint_C \frac{(1+z)\sin z}{(2z-1)^2} dz, \quad C: |z-i|=2 \text{ counterclockwise}$$

Solution